

Rice Science Café

HOTTEST THE THING IN THE UNIVERSE

TUES OCT 4 | 6:30–7:30 pm

Black Labrador Pub, Churchill Room
4100 Montrose Blvd.

FRANK GEURTS, PHD

Associate Professor of Physics & Astronomy
Rice University



RICE | NATURAL SCIENCES

naturalsciences.rice.edu

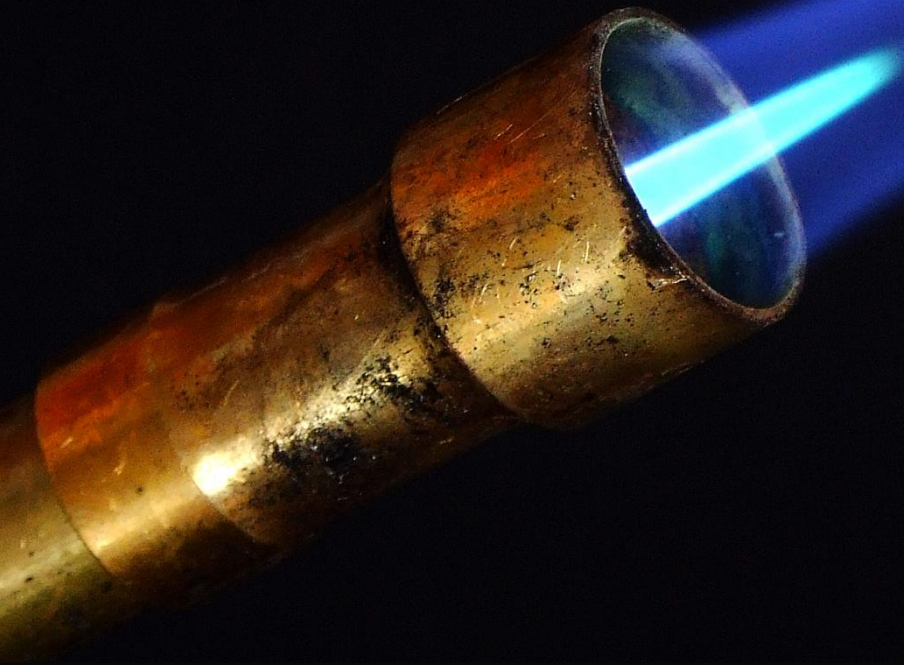
HOT ...

$1000^{\circ} - 2000^{\circ} \text{ F}$



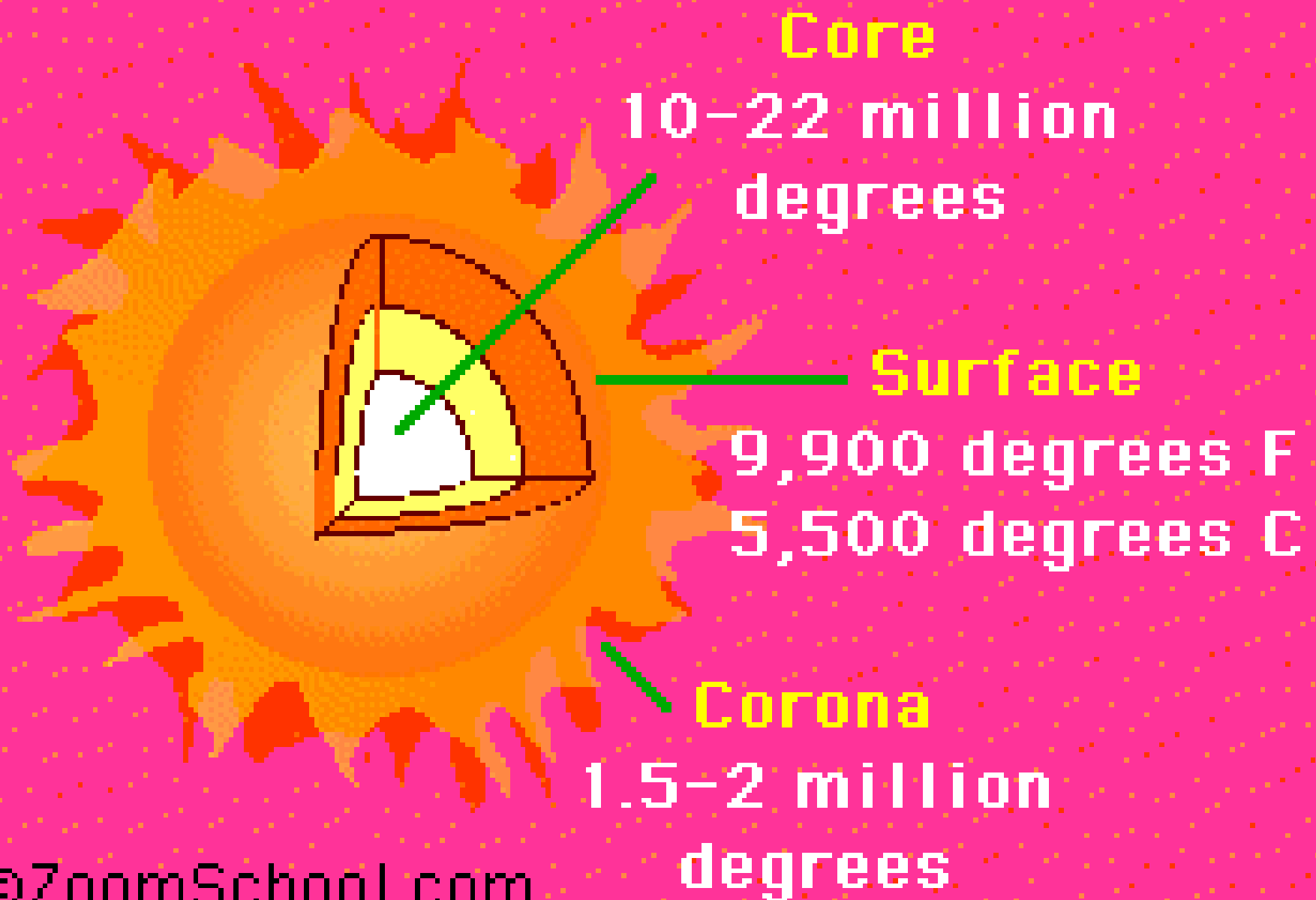
Image Source: wikipedia

2000° – 4000° F



HOTTER ...

Solar Temperatures



HOTTEST !



Back to Earth

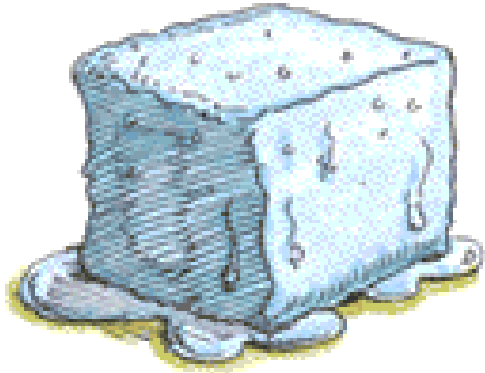
What happens to matter
when you heat it up to
high temperatures?



frozen ... liquid ... vapor ...

remember the states of matter?

**C
O
L
D**



SOLID



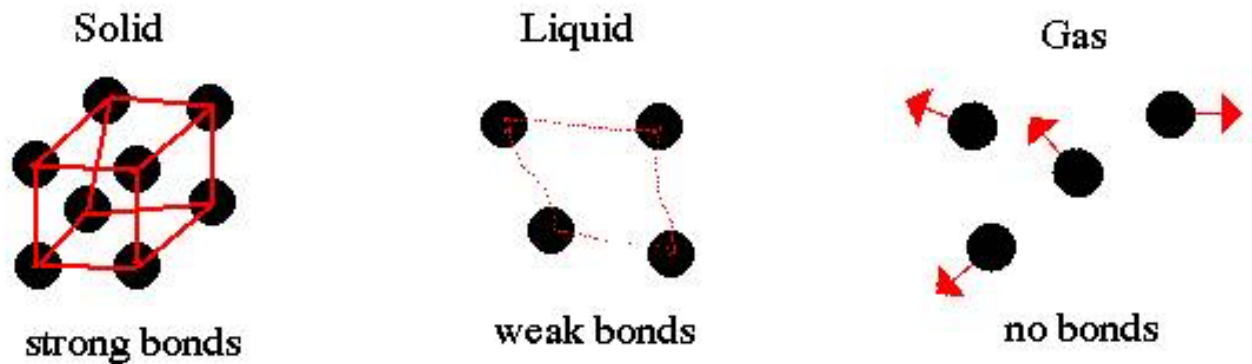
LIQUID



GAS

**H
O
T**

This you learned at school ...



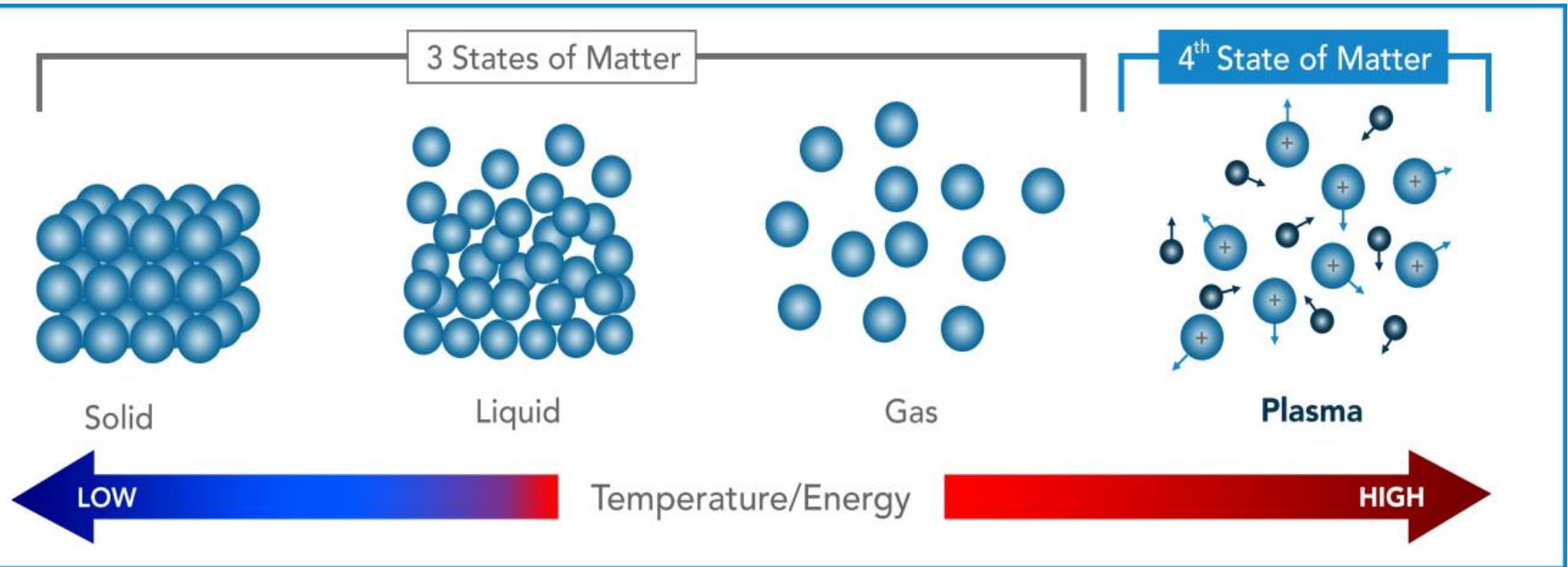
Solid: molecules move very slowly; forces between molecules strong enough to hold them in place.

Liquid: molecules have much more energy; interactions between molecules remains but less dominant.

Gas: interaction between molecules becomes negligible.

Heat it up even more!

... beyond the familiar 3 phases



Molecules start to break up into atoms
Atoms can no longer hold on to its electrons

PLASMA

We are not done yet!

Increase the temperature to extraordinary
temperatures that today can only be achieved
at particle accelerators ...

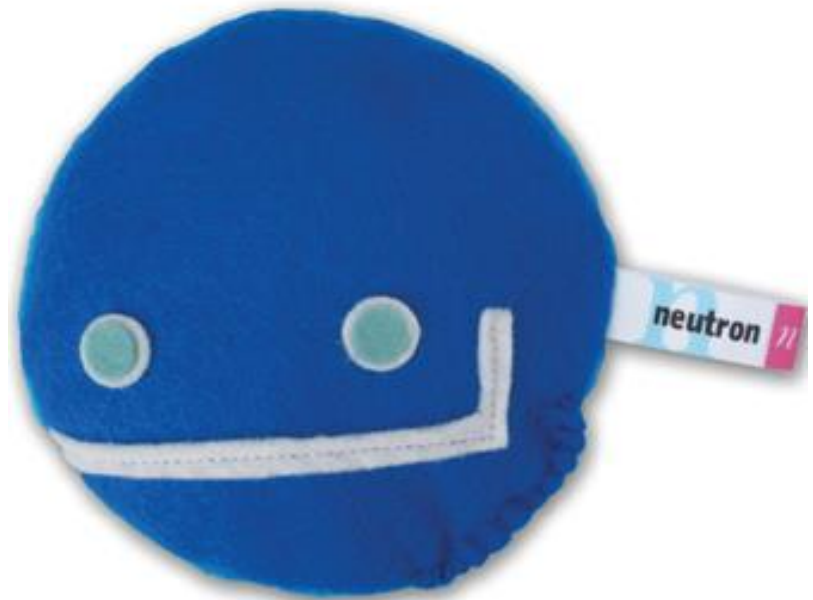


As temperature increases ... nuclei can no longer hold on to its nucleons

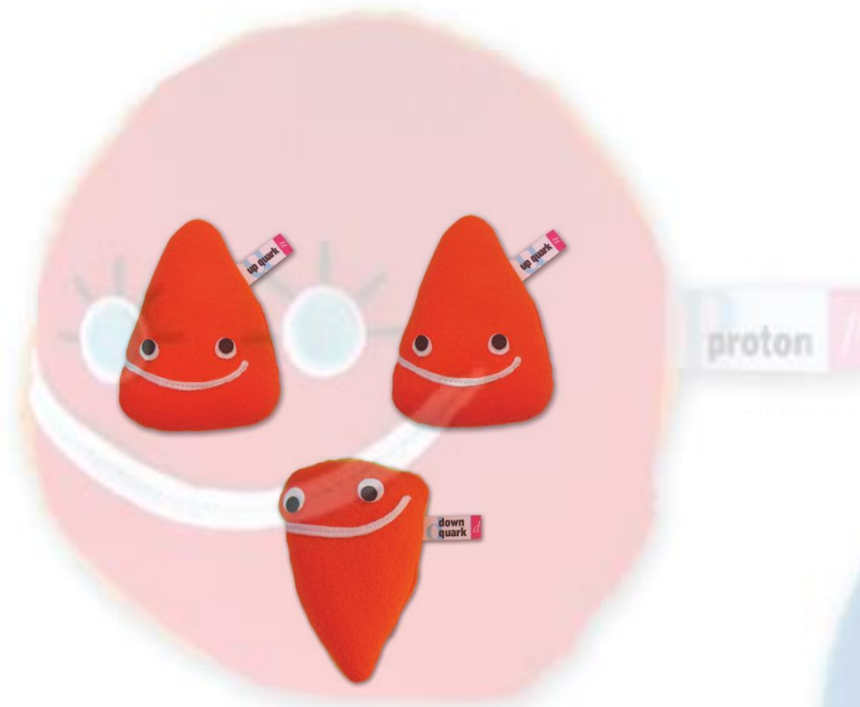
PROTON



NEUTRON

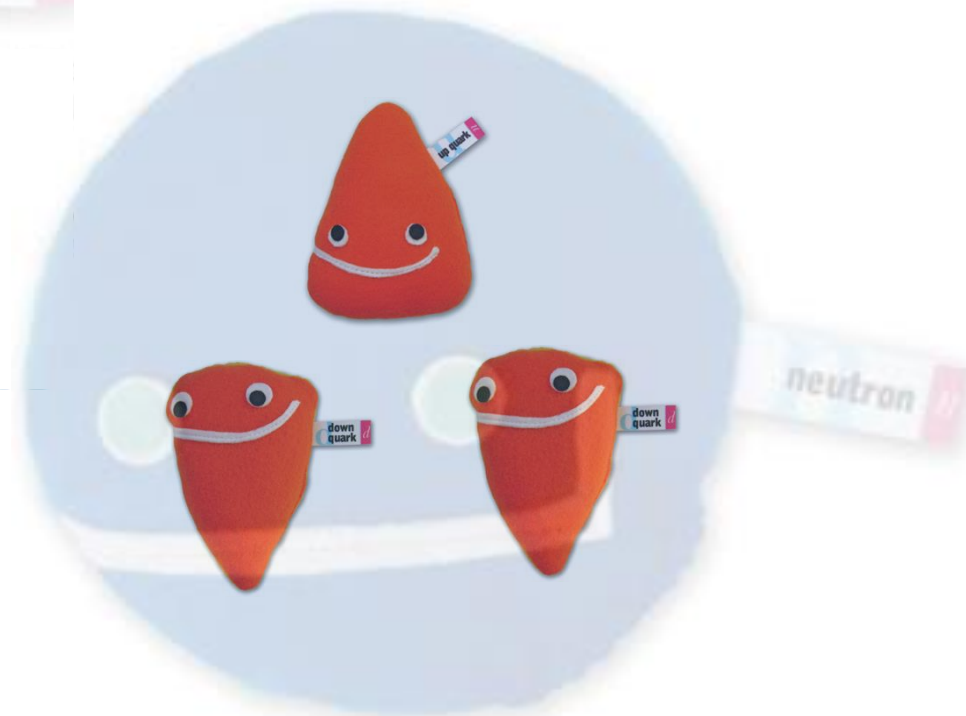


nucleons are each made of 3 quarks



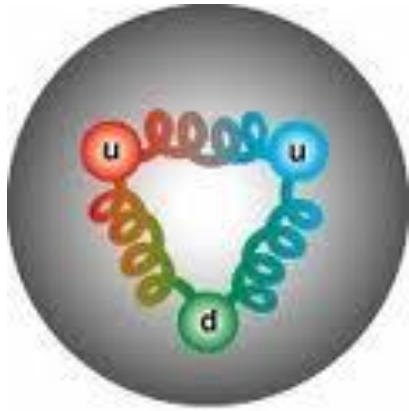
proton = uud

nucleons are *baryons*: qqq or $\bar{q}\bar{q}\bar{q}$
there are also *mesons*: $q\bar{q}$



neutron = udd

Quarks are held together by the **Strong Force**



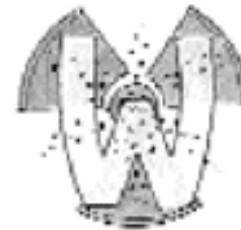
Under normal conditions cannot pull quarks from a nucleon. If you pull hard enough a quark-antiquark pair will be created ... much like a rubber band!



gravity



strong force

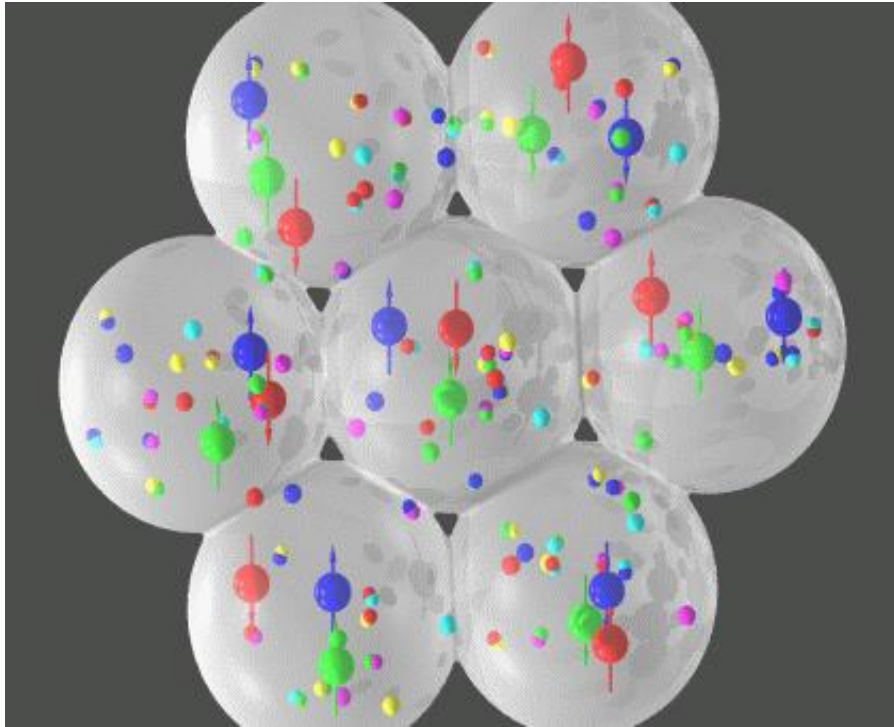


weak force

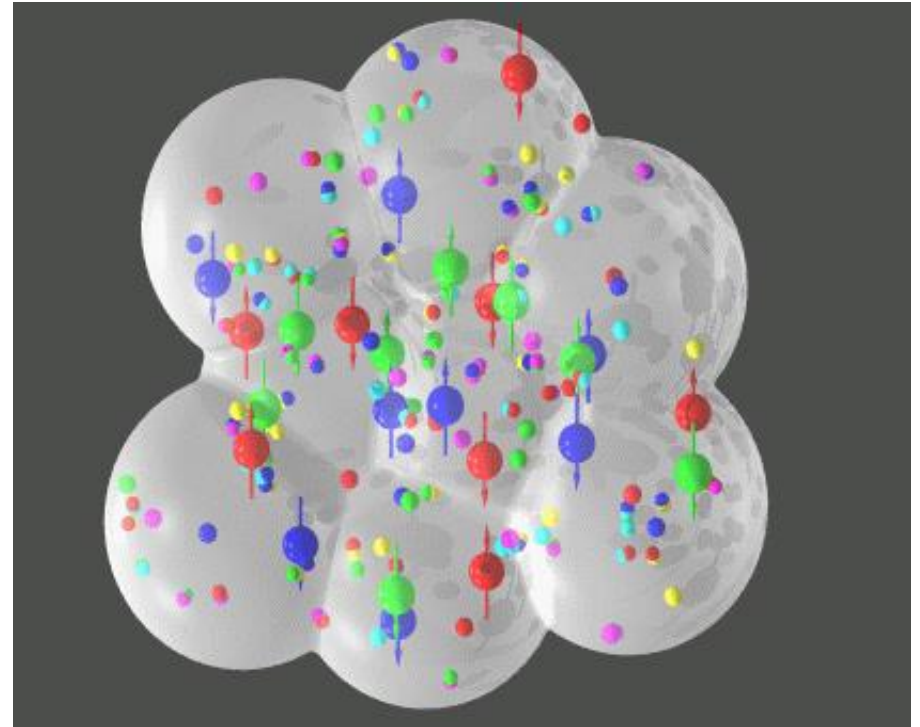


electromagnetism

At particle accelerators, **temperatures** are ANYTHING but normal: nucleons “melt” into quarks and gluons (carrier of the strong force)



Gas on Nucleons

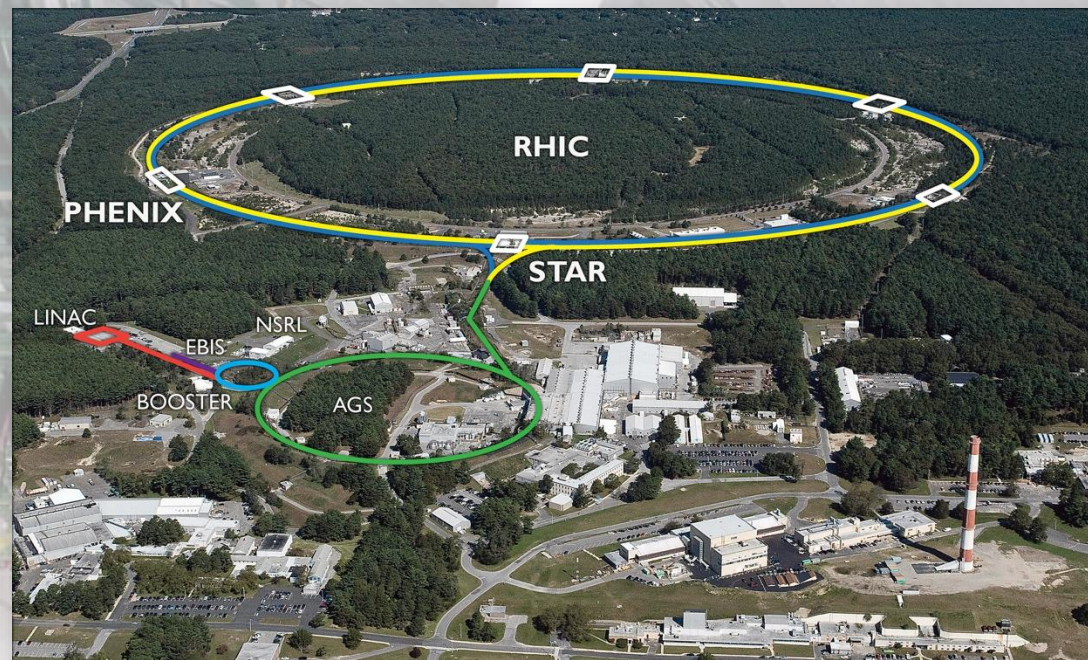


Quark Gluon Plasma

Can we really measure this ...?

RHIC

Relativistic
Heavy Ion
Collider



LHC

Large Hadron Collider

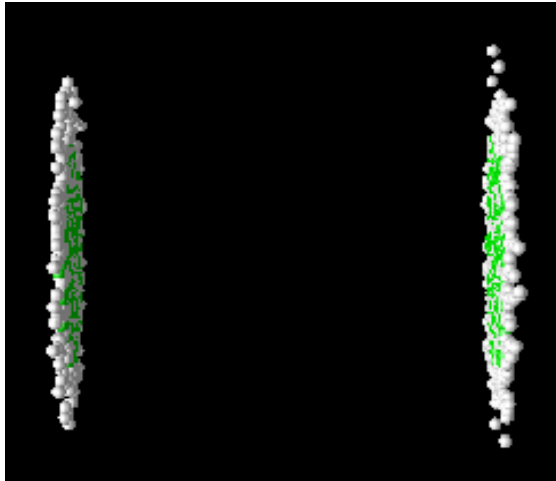


The Recipe:

Accelerate & Collide Heavy Ions

- Take the projectiles: heavy ions
 - large atoms stripped from all its electrons
 - usual candidates are Gold (^{197}Au) and Lead (^{208}Pb)
 - Don't worry: RHIC only uses less than microgram of gold
- Accelerate to large velocities: ultrarelativistic
 - 99.995% of the speed of light (that's 186,000m/s, or 80,000 trips around the RHIC ring every second)
 - Einstein's relativity kicks in: in our lab heavy ions look like flat pancakes!
- And let them collide: dump a lot of energy in a small volume
 - The ions are so small that the force of their impact is about the same as the impact of two mosquitos colliding
 - But, packing so much energy in such a small space is like stuffing all energy consumed in the US for one year into a cube with sides measuring one quarter the width of a single human hair

Ready, Steady, Cook!



Large Particle Accelerators



smash beams
of subatomic particles
at almost the speed
of light ...

**Highest man-made
temperature**

Who

**CERN, LARGE HADRON
COLLIDER**

What

**5×10^{12} DEGREE(S)
KELVIN**

Where

SWITZERLAND

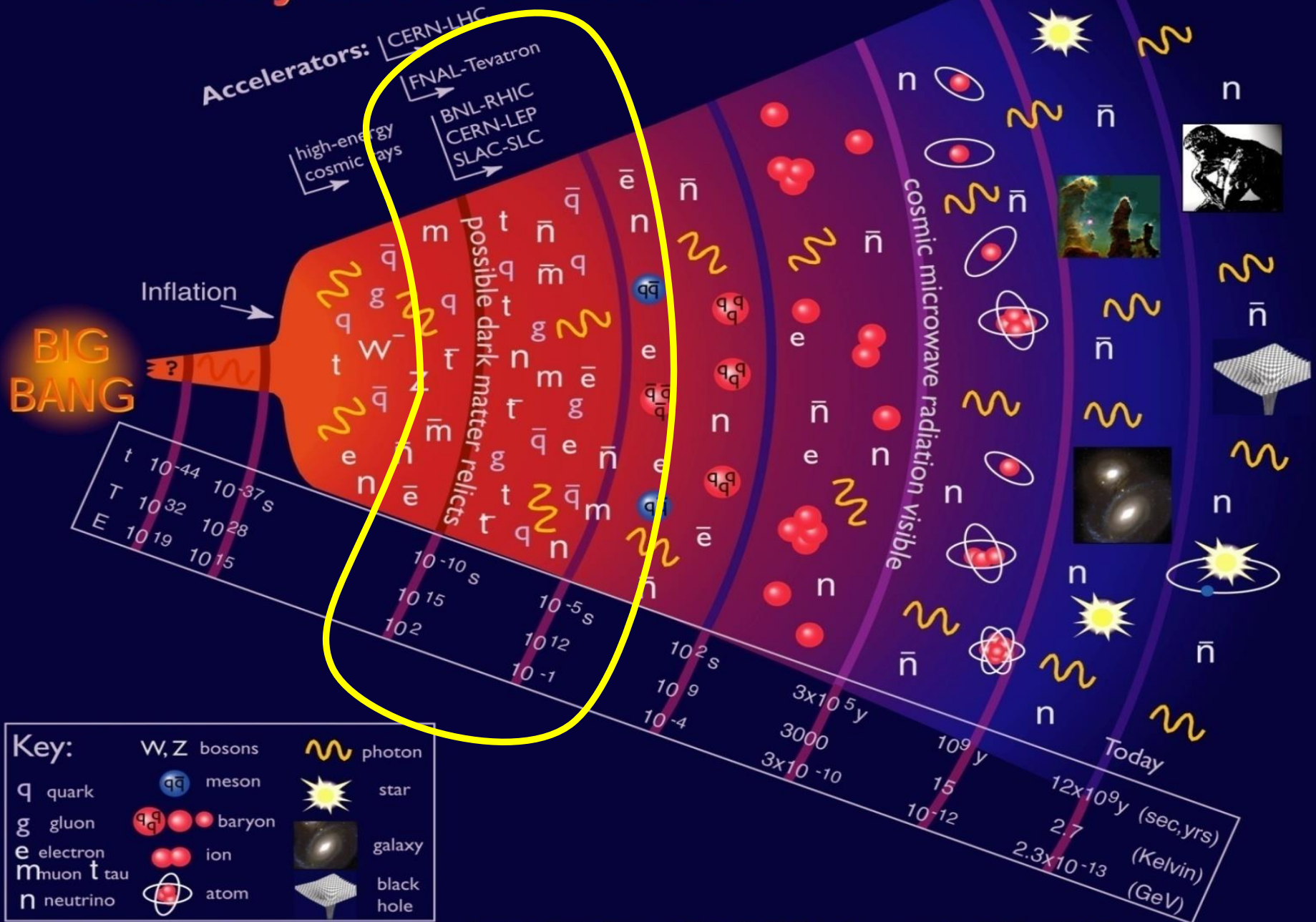
When

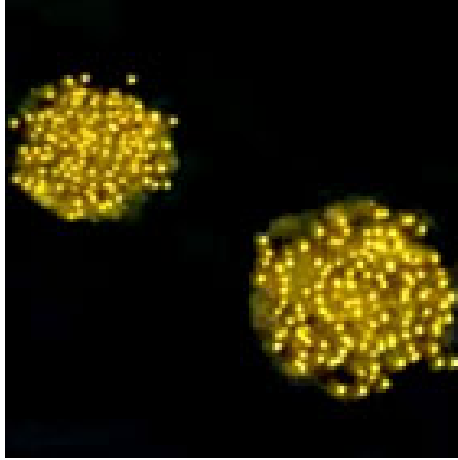
13 AUGUST 2012

Previous record holder:
RHIC at ~4 trillion Kelvin

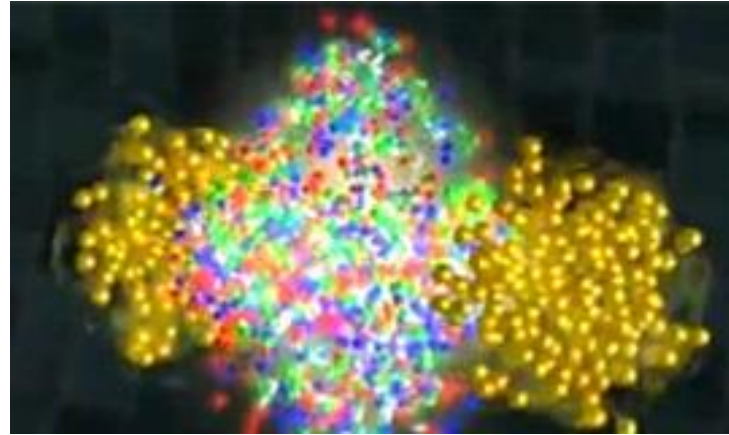
$5 \times 10^{12} \text{ K} \approx 9,000,000,000,000^\circ \text{ F}$

History of the Universe

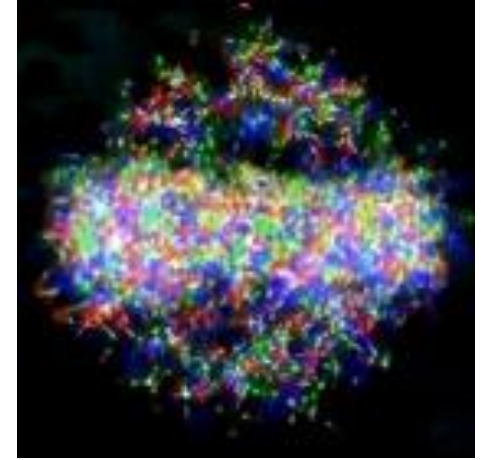




IONS ABOUT
TO COLLIDE



COLLISSION

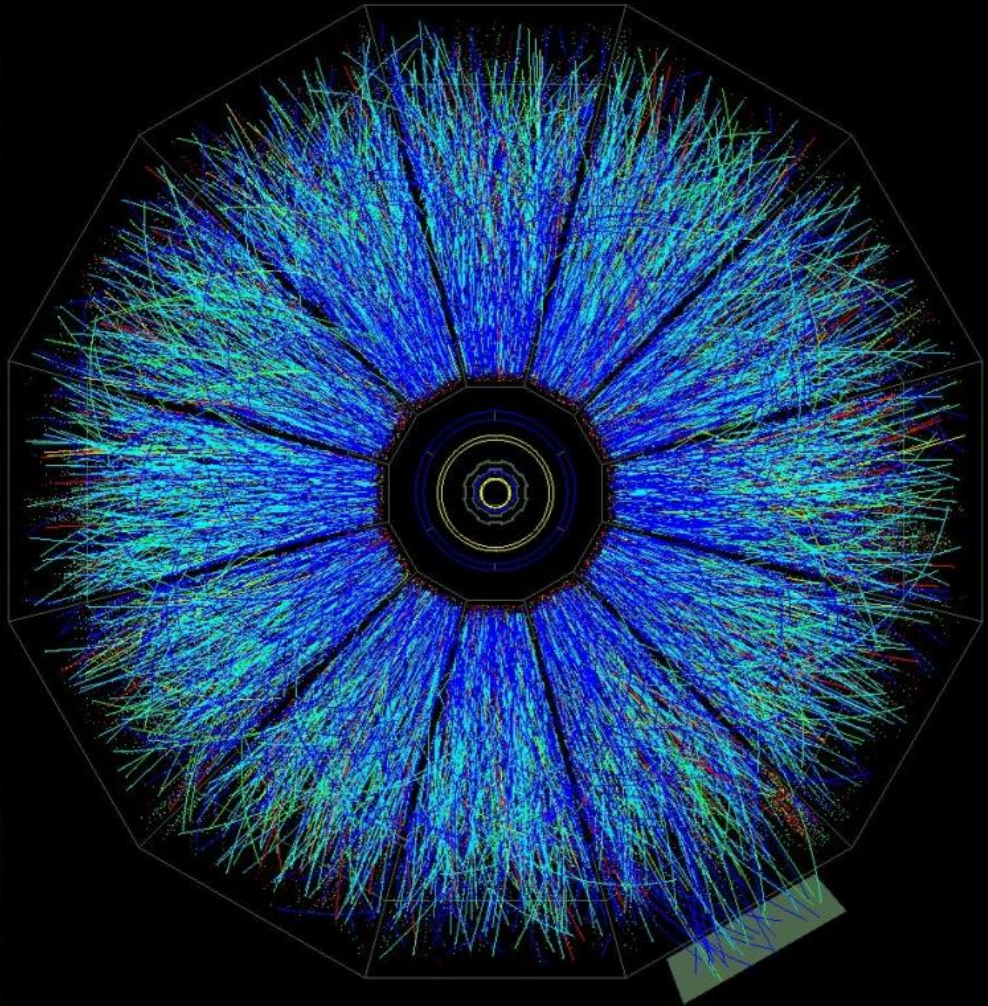


JUST AFER
COLLISSION

Image src: Science & BNL

What have we learned from a
Quark Gluon Plasma ...?

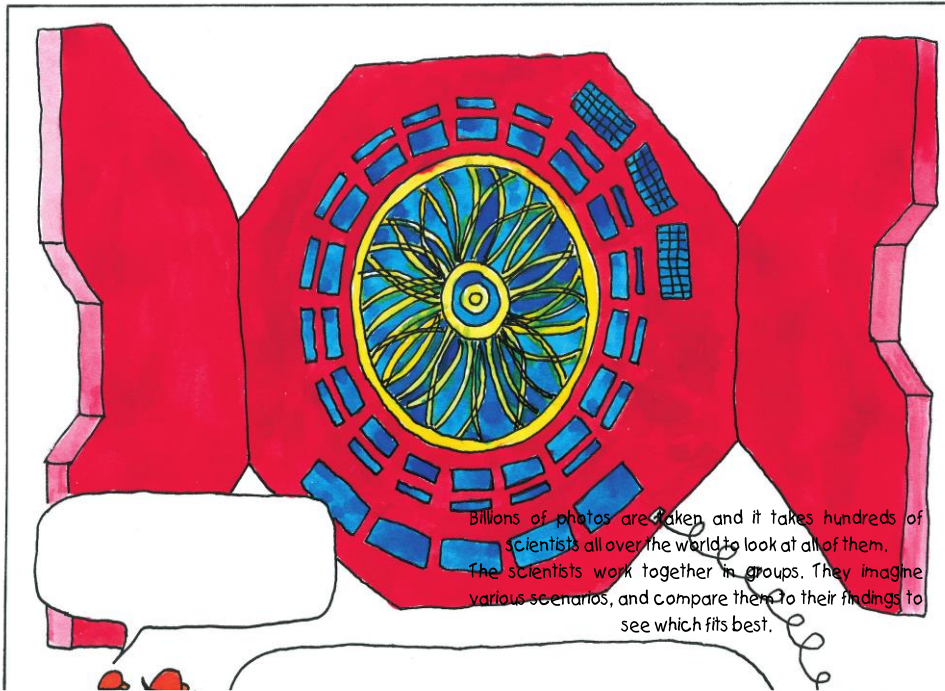
Real collision event display
from a collision of Au+Au



Brockman Hall for Physics

Collect many more collisions ...

Billions of so-called events are collected and digitally stored



We look at all the data and measure:

- “3D” pictures of every collision
- count number particles
- determine the momentum of each particle
- identify the particle (pion, kaon, proton, *etc.*)

This involves large computing farms and many scientists all looking at the different aspects

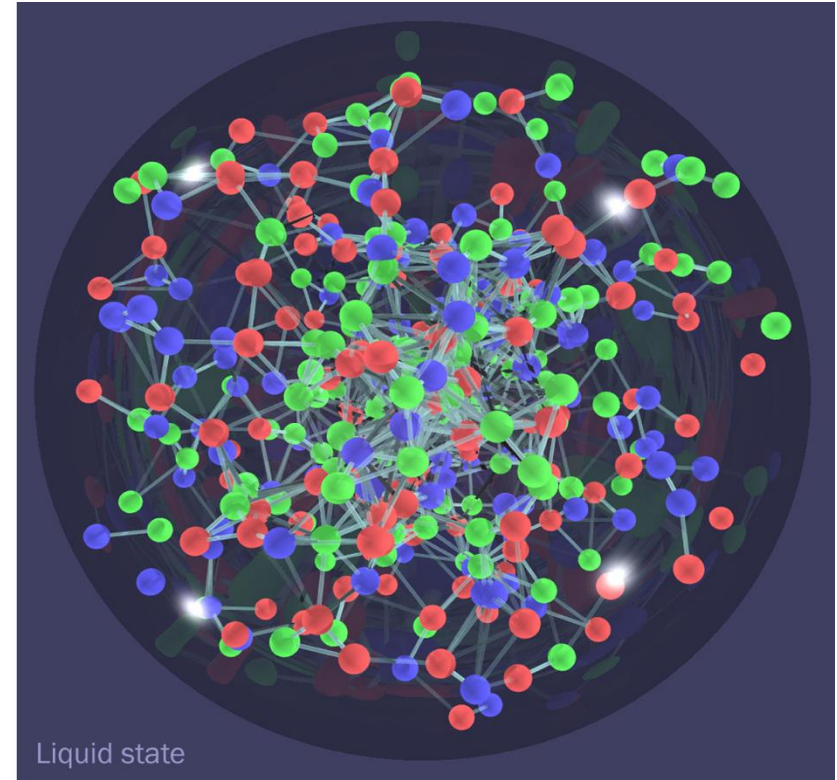
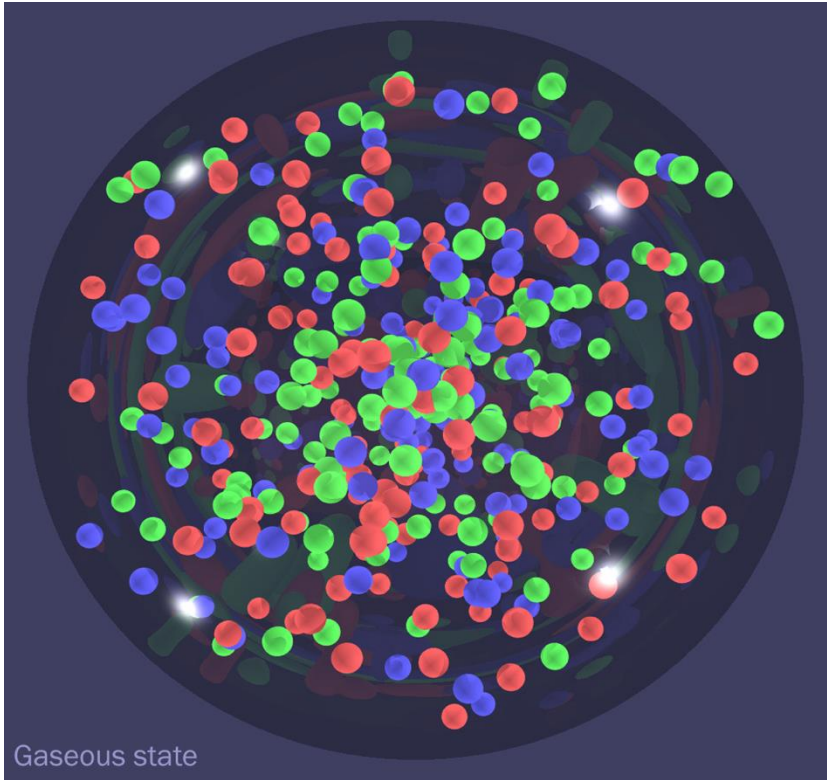
INSIDE THE RECORD-SETTING 7.2 TRILLION-DEGREE COLLISION
CREATED AT BROOKHAVEN NATIONAL LAB...



IT'S NOT
THE HEAT,
IT'S THE
HUMIDITY...

MIT
HARVARD
2012
news-fall

Ok, maybe not the humidity, but ...



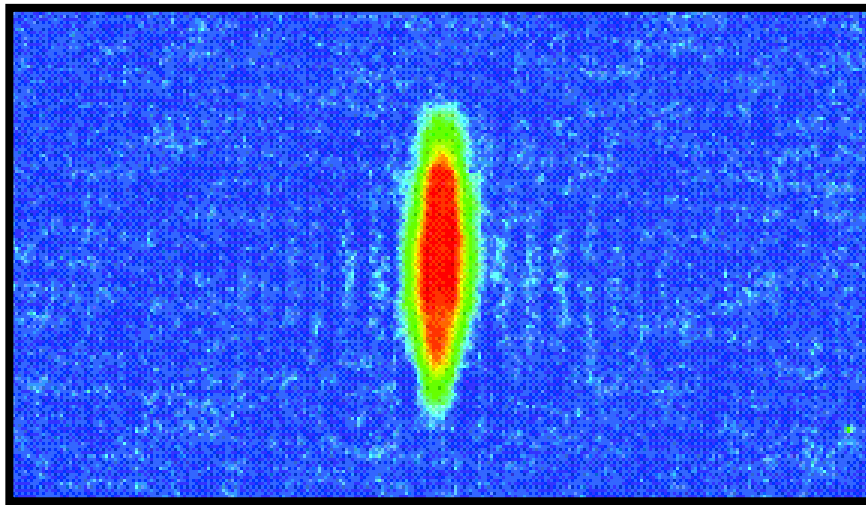
This early-universe matter behaves like a PERFECT liquid
... and not like a gas.

We're Not The Only Ones ...

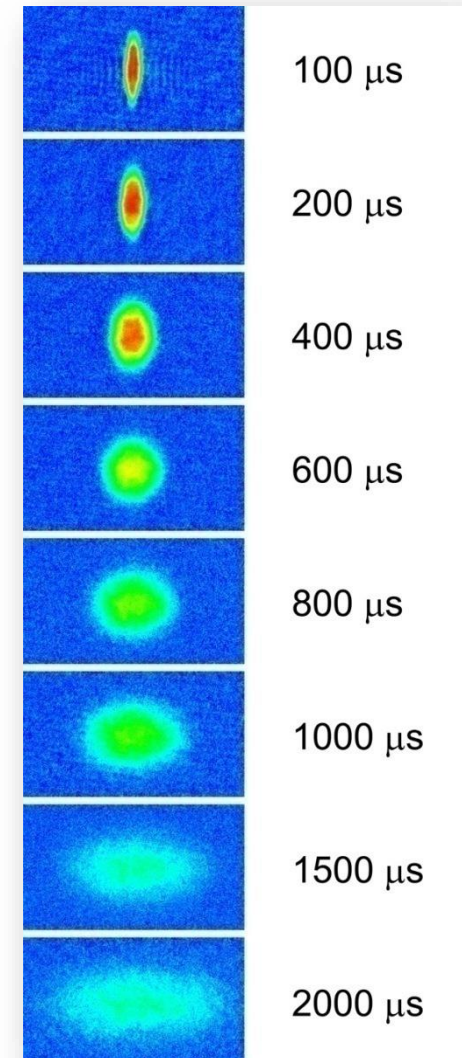
... creating perfect liquids:

similar phenomena in a gas of strongly interacting Li atoms

- difference: ultra-cold!
- commonality: strongly-coupled systems



Science 298 (2002) 2179



What else are we trying to learn from this Quark Gluon Plasma?

STRANGE QUARK



●●●●●●●●○○○○○
LIGHT HEAVY

CHARM QUARK



●●●●●●●●○○○○○
LIGHT HEAVY

BOTTOM QUARK



●●●●●●●●○○○○○
LIGHT HEAVY

a.k.a. "beauty"

TOP QUARK

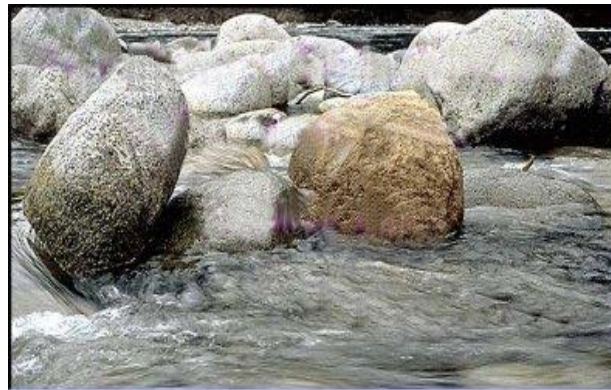


●●●●●●●●○○○○○
LIGHT HEAVY

175x heavier than proton
but is part of the family

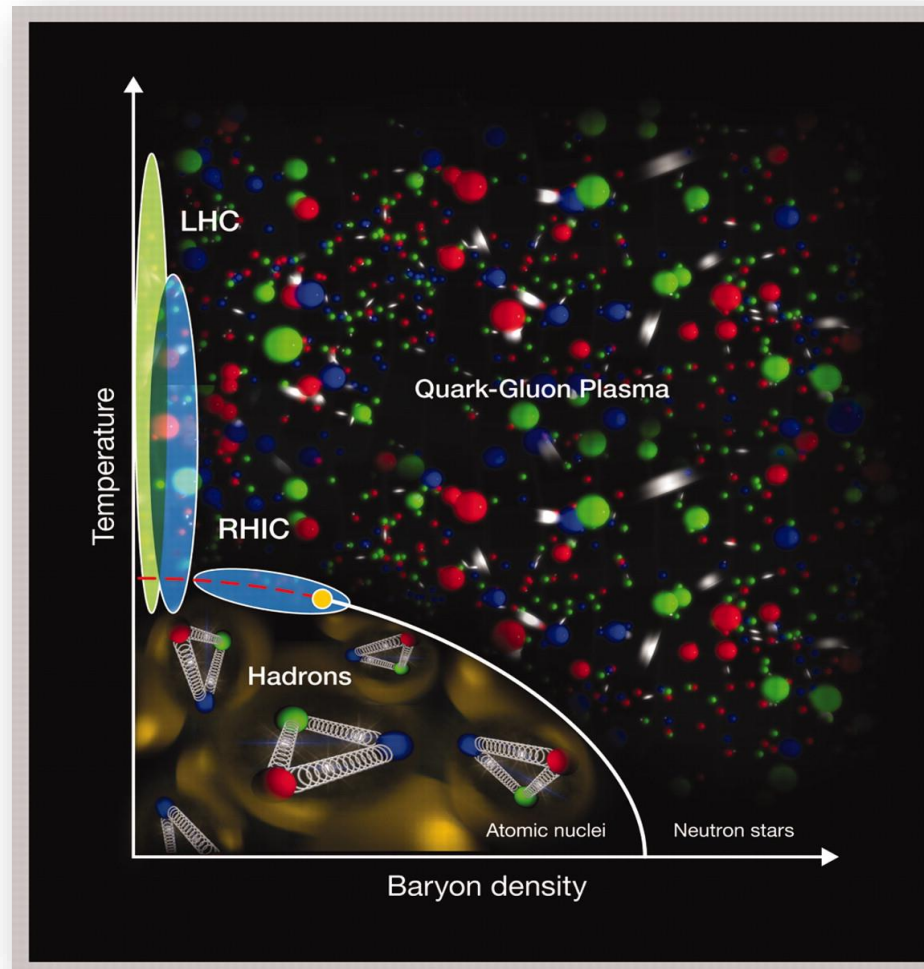
Besides "up" and "down" how do the other (heavier) quarks interact?

Do they "feel" the flow ...



Explore the phase diagram of nuclear matter ...

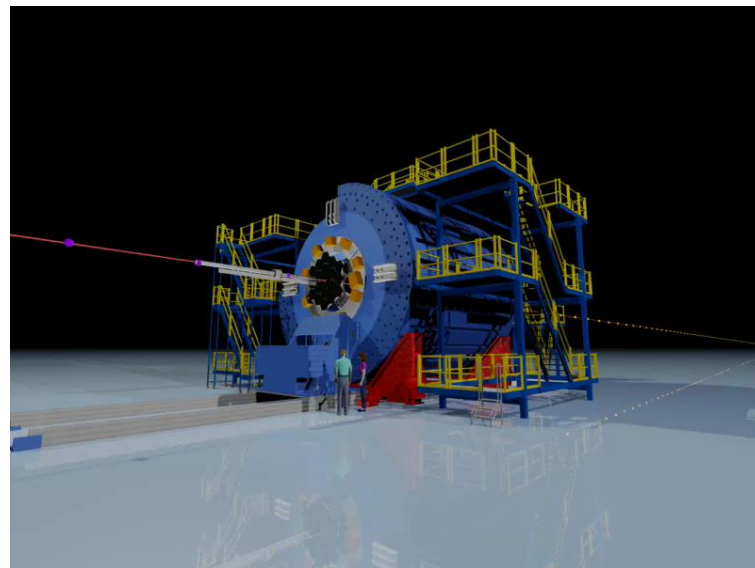
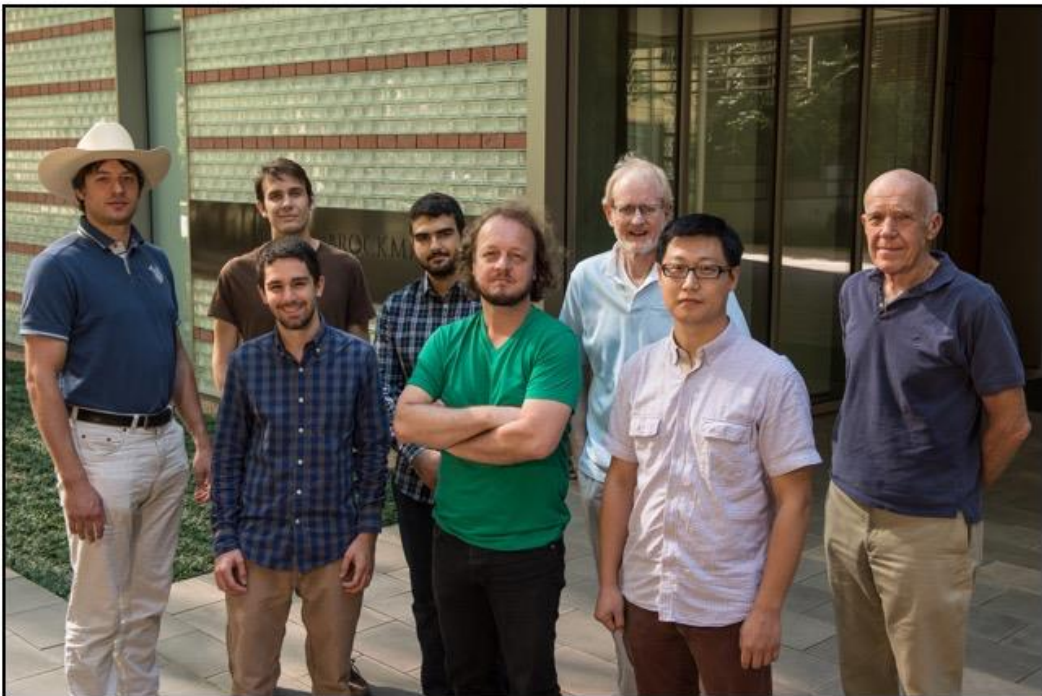
can we find similar landmark signs that are common to other familiar phase diagrams?



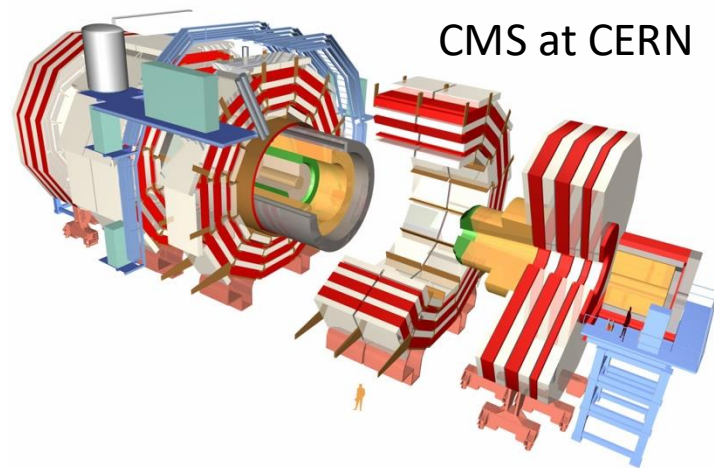
Ultra-Relativistic Heavy Ion Groups at Rice University

<http://heavyions.rice.edu>

STAR at BNL



CMS at CERN



<http://wl33.web.rice.edu>

